
Big Game: Goldman Sach's Elephant Hunt in Libya

Part 1: Options Basics



**Bloomberg
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October 3 – October 9, 2016 | bloomberg.com

Investor of the Decade



With a little help from Goldman Sachs post

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A trip to the Tripoli casino

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Q. How do you tell the \$65bn sovereign wealth fund of a notoriously erratic government that it might lose all its money on trades you arranged for them?

A. Very carefully.

Roadmap

- The situation
- The first Citigroup trade:
 - Mechanics
 - Properties
 - Rationales
- Value bounds:
 - Listed options comparison
 - Put-Call Parity (PCP)

The Situation

- The LIA
- Goldman Sachs
- The elephant trades

The Players



- Founded: 2006
- AuM: \$60bn (2008)
- $\approx$ 20 inexperienced Libyan employees
- CEO: Mohammed Layas
- Deputy CEO: Mustafa Zarti
- Teams: “Equities” and “Alternatives”
- Mission: Large, quick returns for State spending (\$3bn/year)



- Founded: 1869
- HQ: New York, USA
- Revenue: \$69bn (2006)
- Market capitalization: \$96bn (2006)
- 26,000 employees (2006)
- Asset management, commercial banking, investment banking, prime brokerage
- Motto: “*Our clients’ interests always come first*”
- Salesman: Youssef Kabbaj

The “Elephant Trades”

Underlying Stock	Structure	Trade Date	Maturity (Yrs)	Number of shares	Deal Value (\$)
1. Citigroup	Forward + Put	25/1/08	3.0	11.2m	\$100m
2 Citigroup	Forward + Put	29/1/08	3.0	11.1m	\$100m
3 EdF	Forward + Put + Short Call	19/2/08	3.0	3.2m	\$73.8m
4 EdF	Forward + Put + Short Call	22/2/08	3.0	3.2m	\$66.0m
5 EdF	Forward + Put + Short Call	27/2/08	3.0	1.2m	\$37.2m
6 Banco Santander	Forward + Put	24/4/08	3.5	36.2m	\$151.0m
7 Allianz	Forward + Put	25/4/08	3.5	1.7m	\$75.8m
8 Eni	Forward + Put	28/4/08	3.5	21.0m	\$151.4m
9 UniCredit	Forward + Put	28/4/08	3.5	166.3m	\$452.7m
				Total	\$1,208m

Trade Mechanics

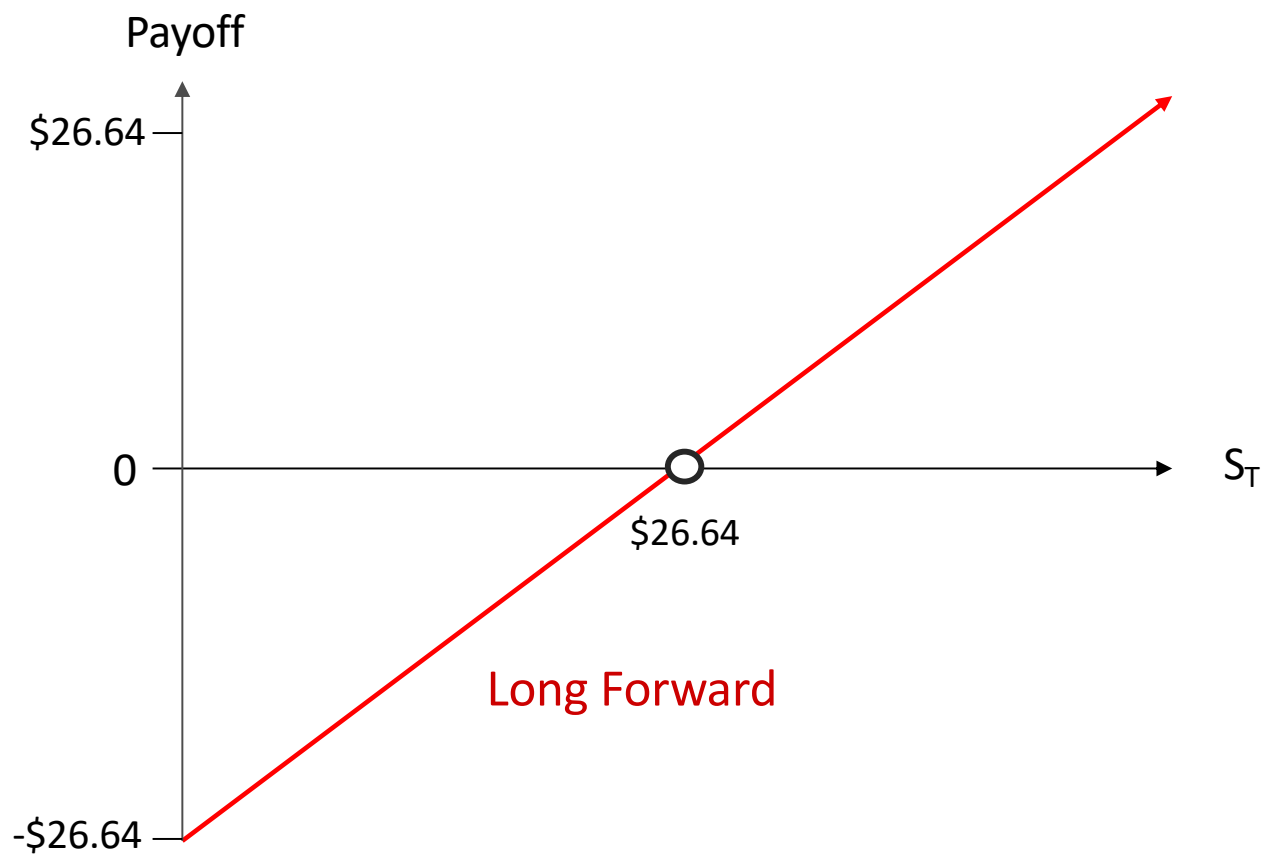
- Long forward
- Long put
- Synthetic position

First Citigroup Trade

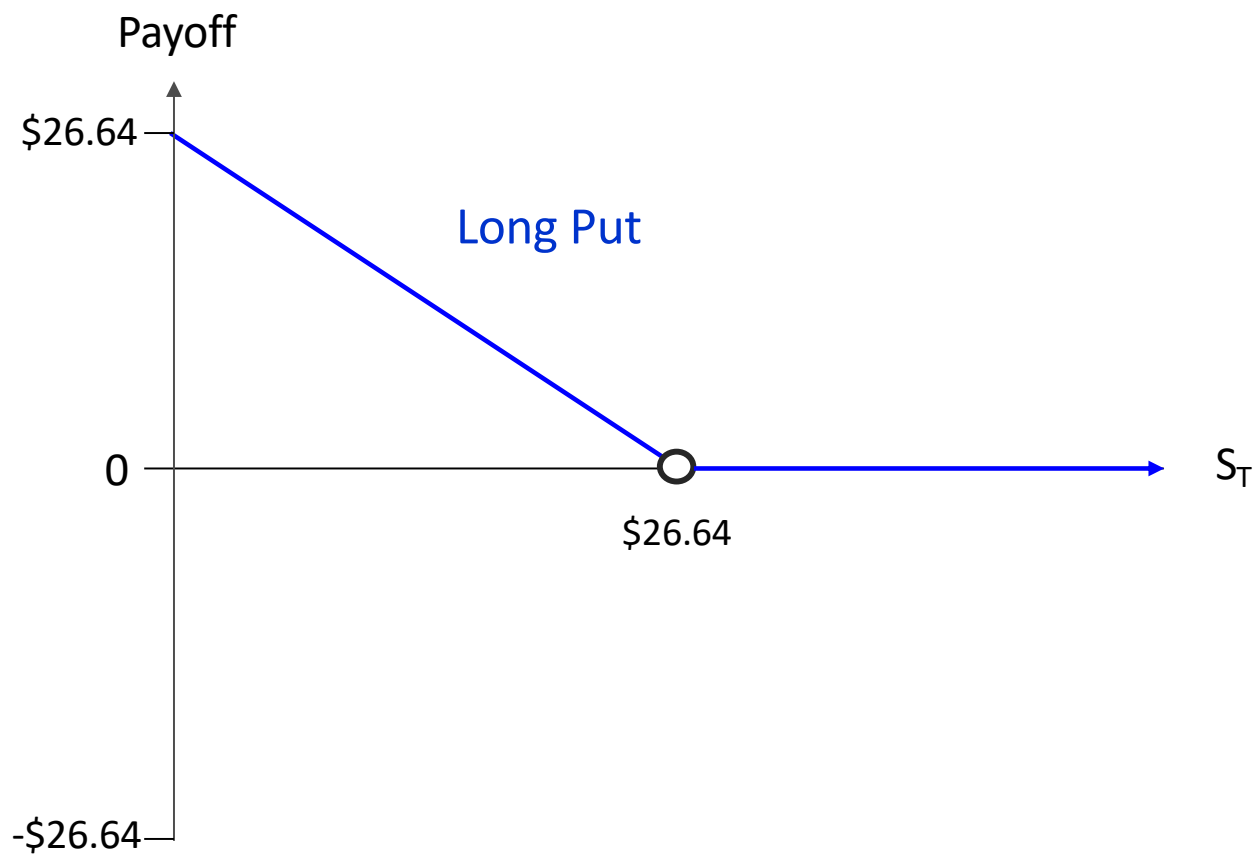
- Trade date: 25/1/2008
- Structure:
 - Forward
 - European put option
 - Both with 3-year maturity (expire 1/21/2011)
 - At-the-money: delivery and strike prices = \$26.64 stock price
- Deal value: \$100m (for 11.2m shares or \$8.92 per share)



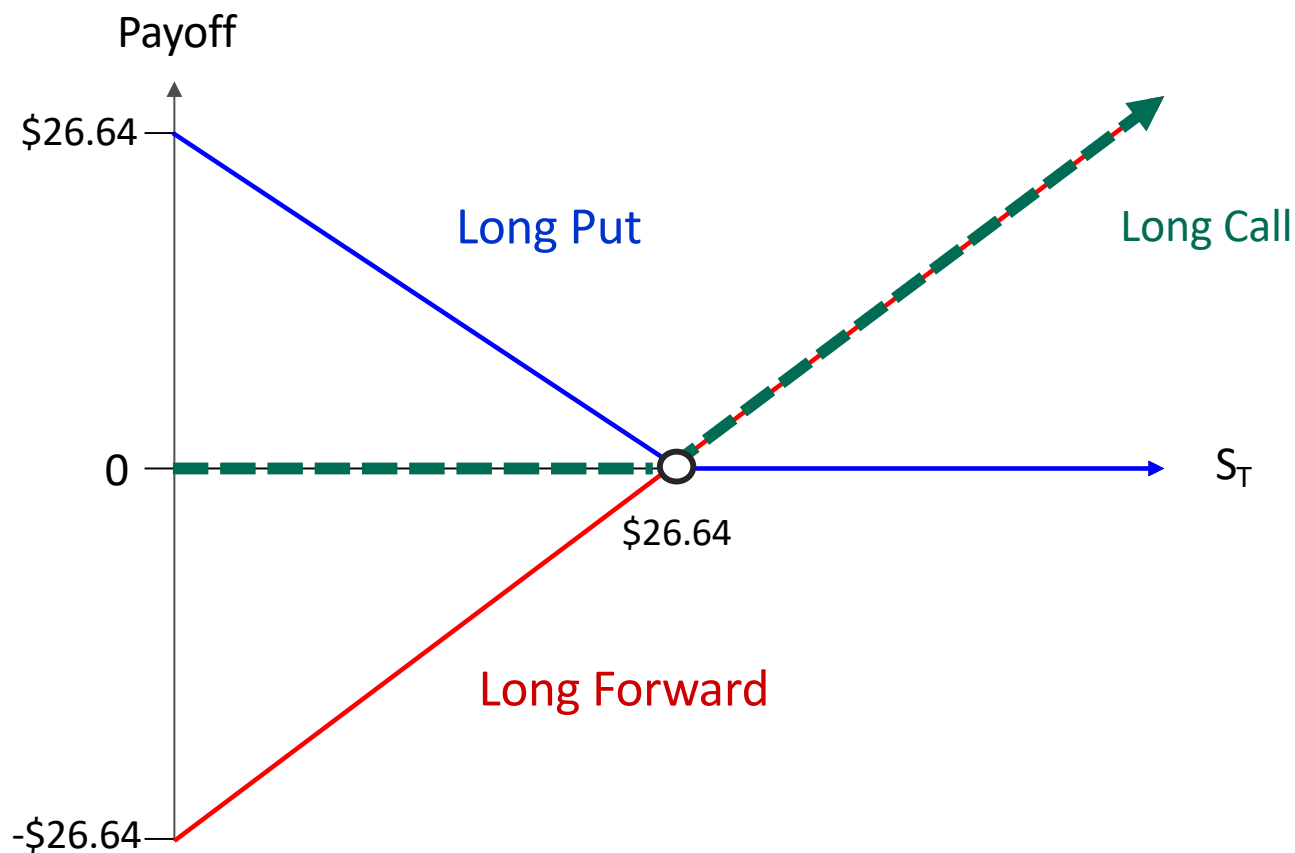
Payoff Diagram: Forward



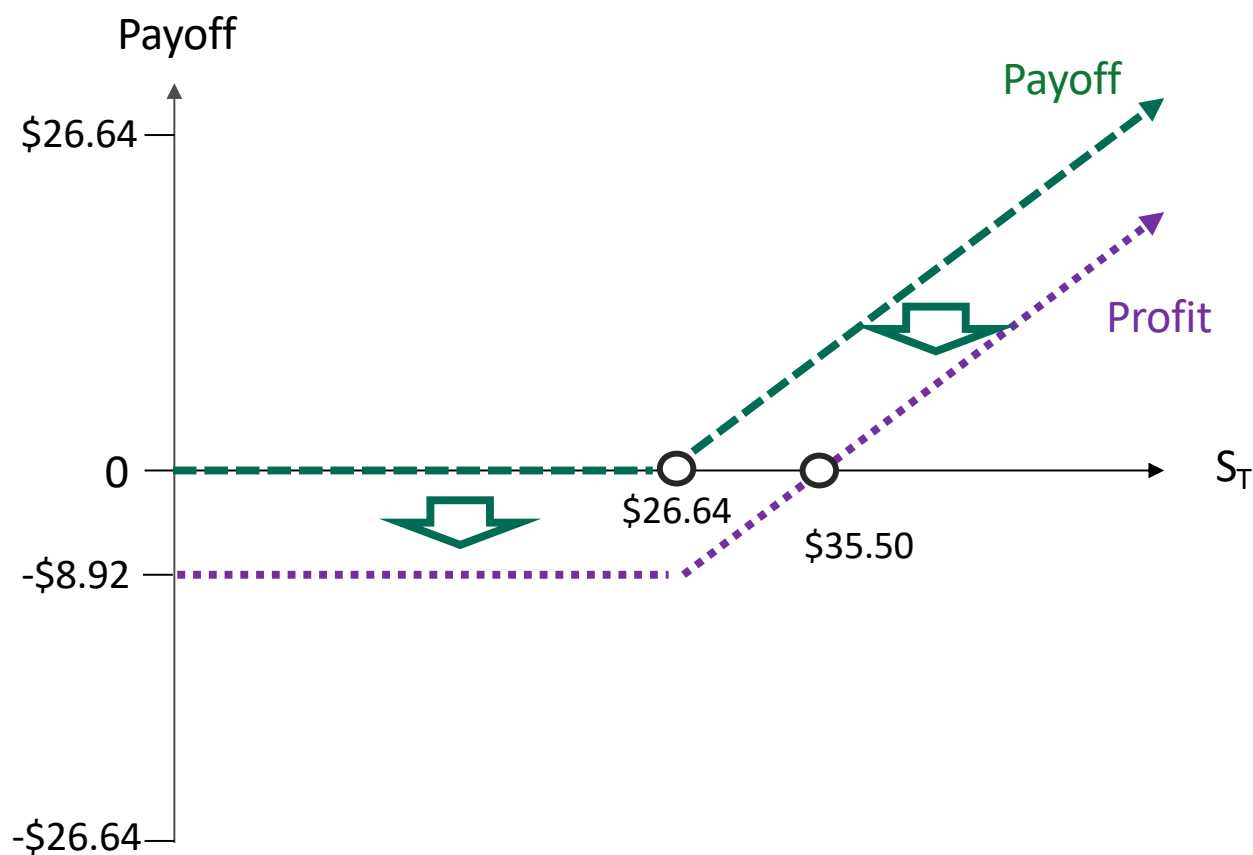
Payoff Diagram: Put Option



Payoff Diagram: Forward + Put = Call Option



From Payoff to Profit



(Profit at T ignores time value of money on the \$100m premium)

Goldman's Scenario Analysis

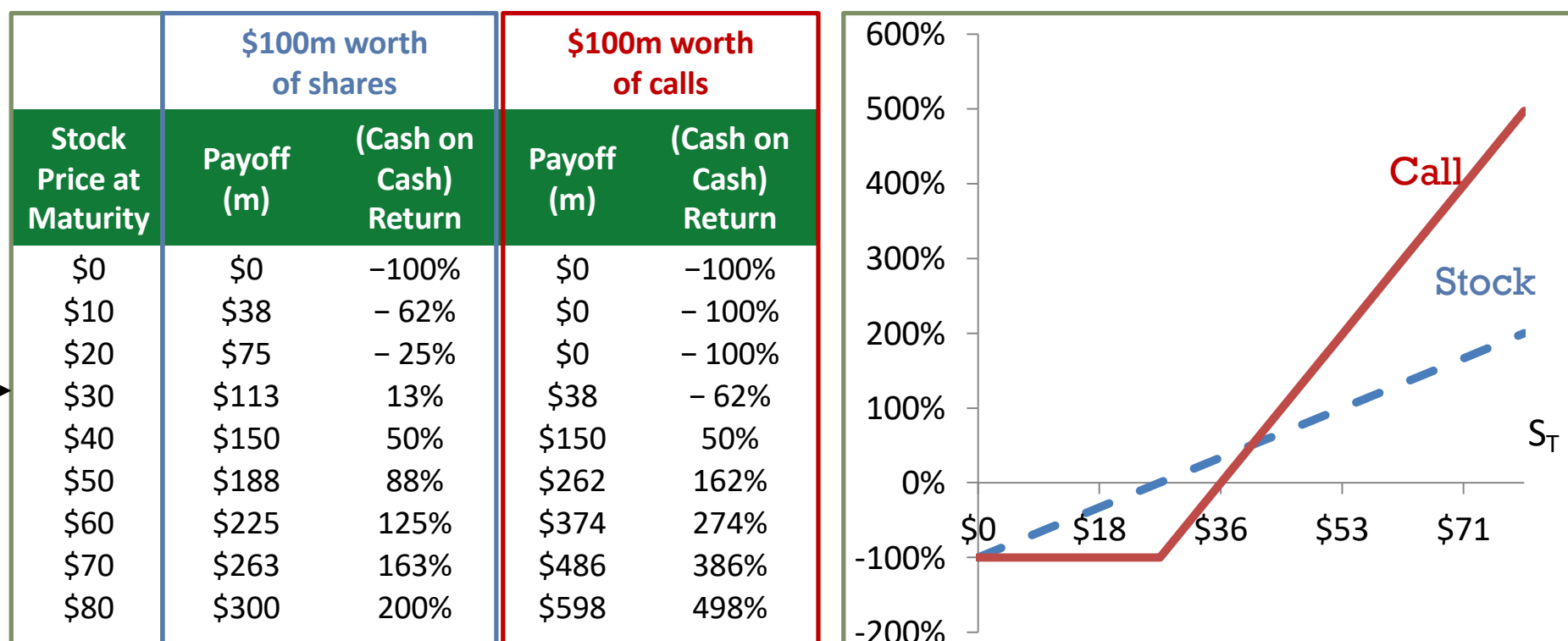
Chigroup Share Price at Maturity	Payoff	Cash on Cash Return	Indicative IRR
\$ 32.00	\$ 54,874,271	-45.33%	-18.23%
\$ 34.00	\$ 77,088,373	-22.91%	-8.31%
\$ 36.00	\$ 99,502,475	-0.50%	-0.17%
\$ 38.00	\$ 121,916,577	21.92%	6.83%
\$ 40.00	\$ 144,330,679	44.33%	13.01%
\$ 42.00	\$ 166,744,781	66.74%	18.58%
\$ 44.00	\$ 189,158,883	89.16%	23.87%
\$ 46.00	\$ 211,572,985	111.57%	28.38%
\$ 48.00	\$ 233,987,087	133.99%	32.78%
\$ 50.00	\$ 256,401,189	156.40%	38.87%
\$ 52.00	\$ 278,815,291	178.82%	40.75%
\$ 54.00	\$ 301,229,393	201.23%	44.42%
\$ 56.00	\$ 323,643,495	223.64%	47.92%
\$ 58.00	\$ 346,057,597	246.06%	51.26%
\$ 60.00	\$ 368,471,699	268.47%	54.45%
\$ 62.00	\$ 390,885,801	290.89%	57.53%
\$ 64.00	\$ 413,299,903	313.30%	60.48%
\$ 66.00	\$ 435,714,005	335.71%	63.33%
\$ 68.00	\$ 458,128,107	358.13%	66.08%
\$ 70.00	\$ 480,542,209	380.54%	68.75%
\$ 72.00	\$ 502,956,311	402.96%	71.33%
\$ 74.00	\$ 525,370,413	425.37%	73.84%
\$ 76.00	\$ 547,784,515	447.78%	76.28%
\$ 78.00	\$ 570,198,617	470.20%	78.65%
\$ 80.00	\$ 592,612,719	492.61%	80.96%
\$ 82.00	\$ 615,026,821	515.03%	83.22%

Trade Properties

- Leverage
- Counterparty risk
- Over-the-Counter (OTC) / Synthetic Trade

Leverage

- Return at maturity (simple compounding) for \$100m invested in:
 - Citigroup shares: $\$100\text{m}/\$26.64 = 3.75\text{m shares}$
 - Citigroup calls: $\$100\text{m}/\$8.92 = 11.2\text{m calls}$

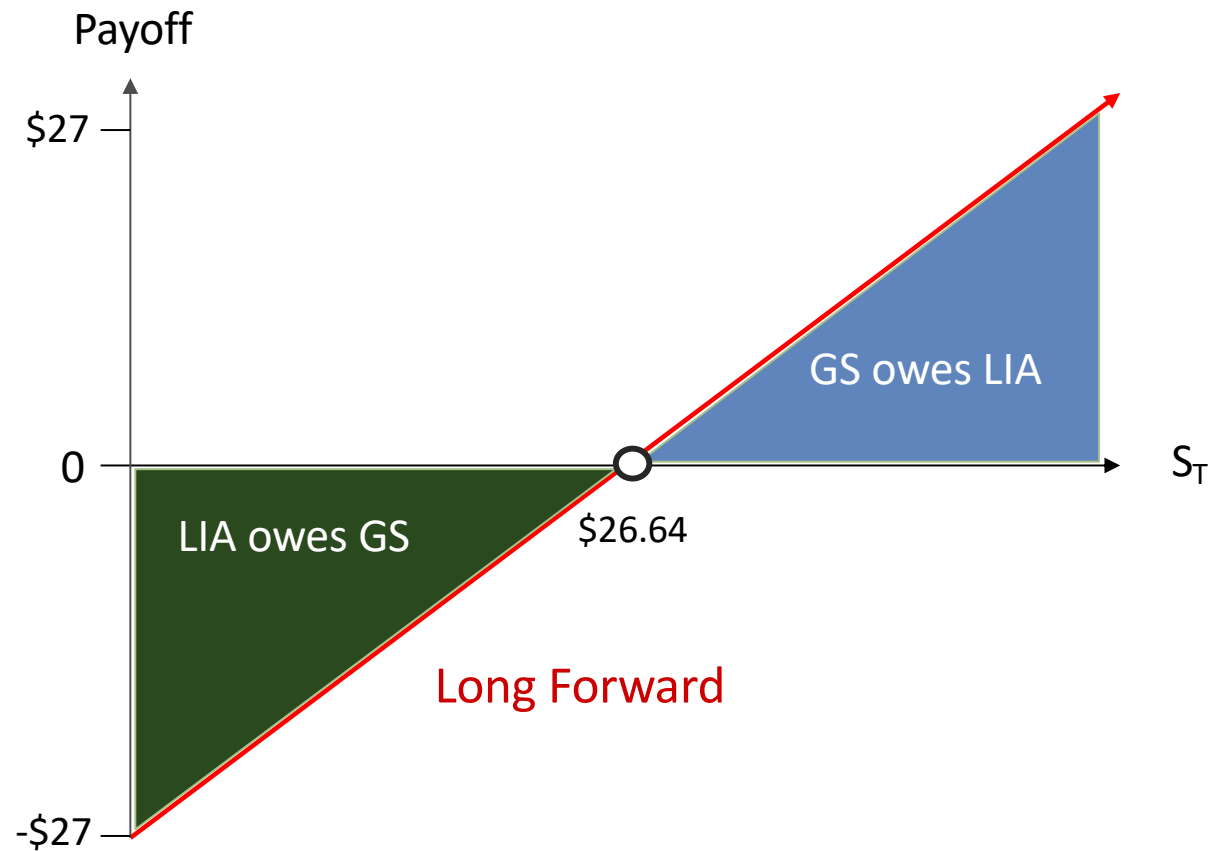


e.g., if $S_T = \$30$, call payoff = $11.2\text{m} \times (\$30 - \$26.64) = \$38\text{m}$ and return = $(38 - 100)/100 = -62\%$

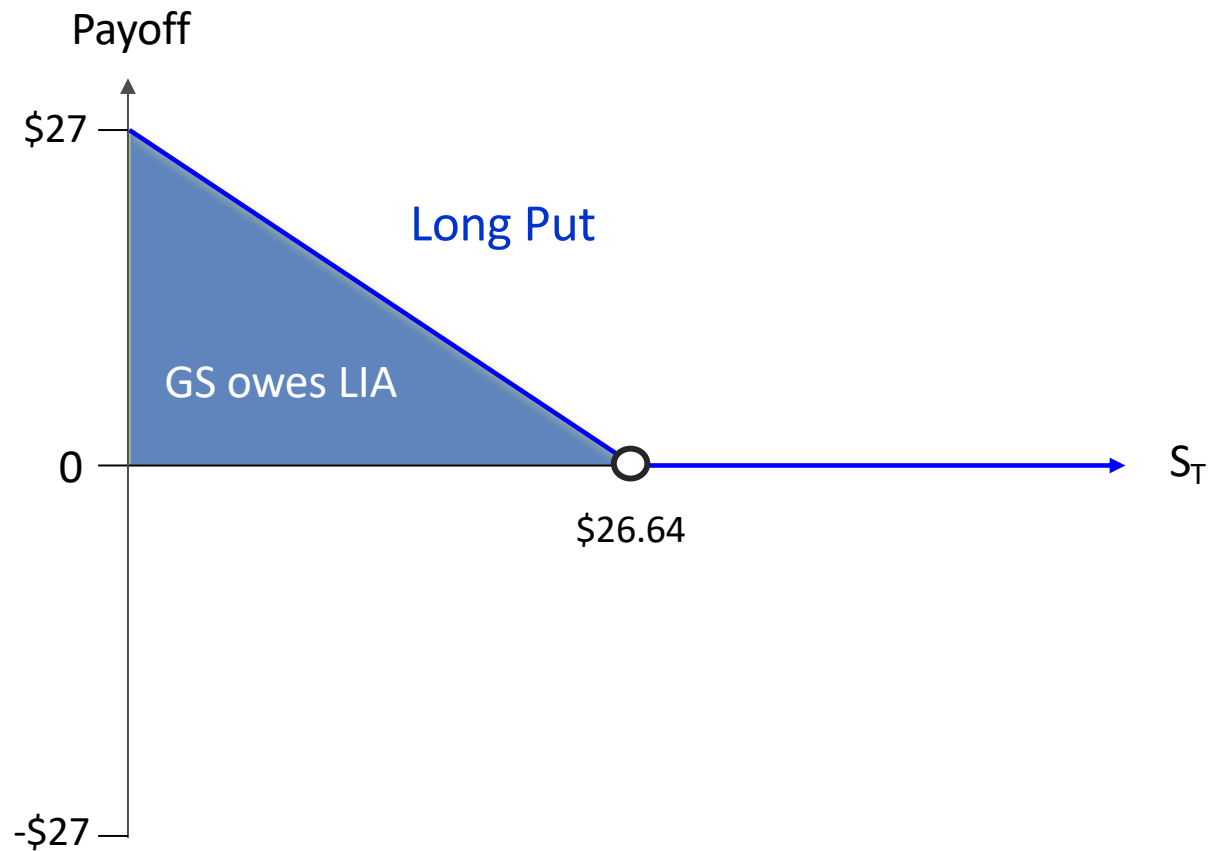
Intuition

- *The call position is riskier than a position in the stock:*
 - Compared to the stock, the call's performance is:
 - + higher if the stock price at maturity is high
 - + lower if the stock price at maturity is low
 - The LIA can even lose everything (if the stock price is below \$26.64)
- *Why?*
 - A call is like the stock plus downside protection
 - If protection were free, the call would be safer than the stock
 - But the LIA must pay for it by buying the put, and that investment is risky: in many cases, it will not pay off

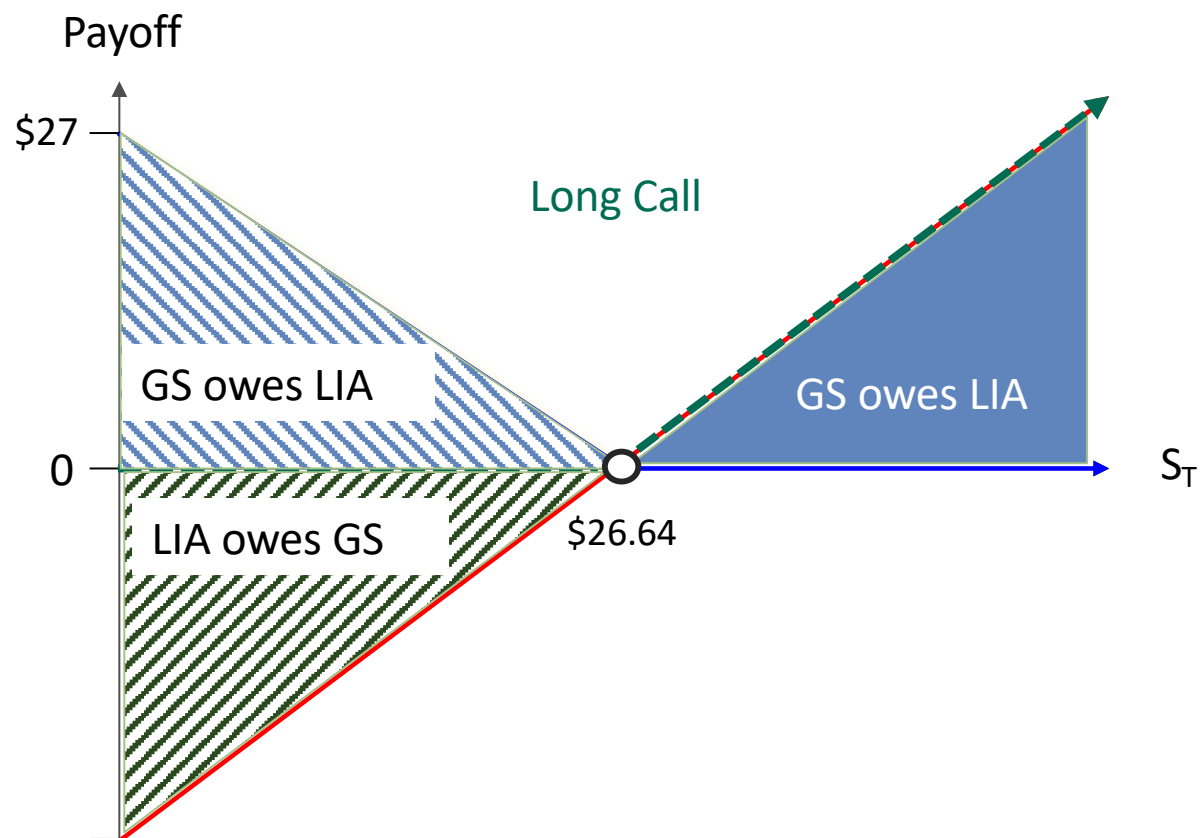
Counterparty Risk: Forward



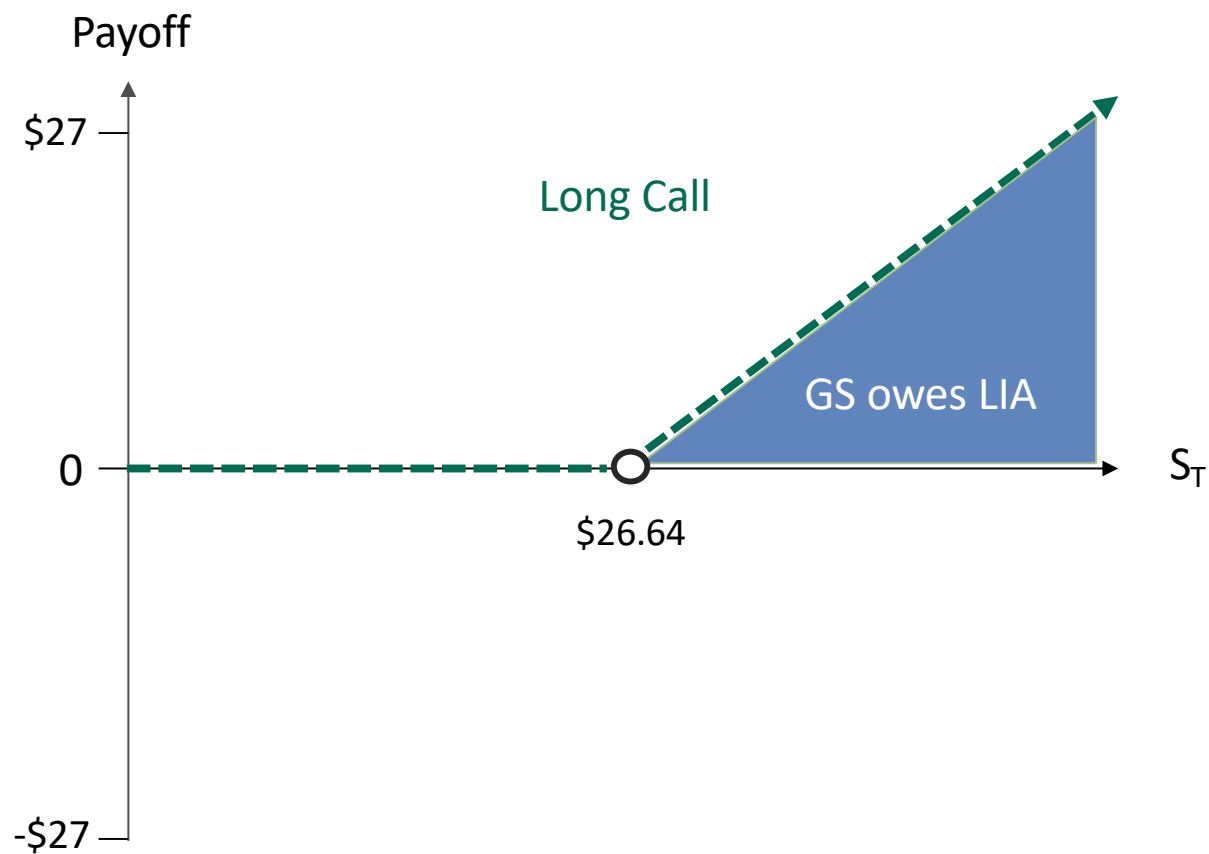
Counterparty Risk: Put



Counterparty Risk: Forward + Put



Counterparty Risk: Call



OTC & Synthetic

- Over-the-Counter (OTC)
 - Bespoke bilateral LIA-Goldman Sachs contract
 - Listed options:
 - + Standardized (e.g. no 3-year maturity listed options on Citigroup)
 - + Option market not liquid enough for such a large trade
- Synthetic
 - No direct stock positions
 - Concern they might be frozen if new sanctions (e.g. Pugh case)
 - NB: The LIA is inconsistent as they bought shares in US companies

Trade Rationale

- The LIA
- Goldman Sachs

The LIA's Rationale

- Speculation (i.e. good deal)?
 - Citigroup (lost half its value in 6 months) is temporarily underpriced
 - Other sovereign funds (Qatar, Abu Dhabi) invested too
 - Kabbaj's SMS: *"It's time to do the trade!!!"*
- Hedging (i.e. to offset other risks)? No
- Mistake?
 - Beware hindsight bias: as of 2007-08, the crisis' scale was less clear
 - Large returns needed to fund State spending → High risk
 - Strategic interest (but 11.2m shares = 0.2% + cash-settled)

Goldman's Rationale

- Goldman's position
 - They take the other side (counterparty)
 - But they hedge that exposure to the extent possible
 - ➔ Not betting against the LIA
- Cost/Benefit
 - Benefit: Premium (\$100m) paid by the LIA
 - Costs:
 - + setting up the trade (e.g. salaries, etc.)
 - + hedging (e.g. the price of positions + transaction costs)
 - + residual risk due to imperfect hedging

How Does Goldman Hedge the Trade?

- Pre-hedging, Goldman is exposed to two main risks:
 1. “Delta”: changes in the stock price (price $\uparrow \rightarrow$ trade $\uparrow \rightarrow$ GS $\downarrow$)
 2. “Vega”: changes in the stock’s volatility (vol. $\uparrow \rightarrow$ trade $\uparrow \rightarrow$ GS $\downarrow$)
- 1. “Delta-hedging”: Buy Citigroup shares, adjusting the amount over the trade’s duration $\rightarrow$ Goldman largely immune to stock price changes
- 2. But hedge of volatility risk is imperfect $\rightarrow$ Large residual risk
 - Trading options... but low volume in markets + no 3-year options
 - Deal with large banks... but limited appetite + would raise prices
 - $\rightarrow$ Traded options on an index correlated with Citigroup (e.g. S&P500)

How is the Premium Set?

1. *Midmarket price* set by trading team using pricing models
2. *Walkaway* added:
 - ➔ *Walkaway price* = minimum acceptable by the bank
 - To set walkaway, make judgement-based adjustments
 - Add “volatility points” in the pricing model (➔ higher trade value) to reflect unhedged risk the bank is exposed to.
3. The sales team negotiates with the client a price above the walkaway price, i.e., it adds a *mark-up*

Value Bounds

- Listed options comparison
- Put-Call Parity

Listed Options Comparison

Expiration Date	Option Symbol	Type	Strike Price (\$)	Bid (\$)	Ask (\$)	Open Interests
17-Jan-09	VRN.AE	Call	25	4.70	4.70	39,297
	VRN.AF	Call	30	2.60	2.61	87,923
16-Jan-10	WRV.AE	Call	25	6.10	6.40	19,821
	WRV.AF	Call	30	4.15	4.45	44,015

25-Jan-1	OTC	Call	26.64	Paid: 8.92	n/a
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Change	Effect on call price
Time to Maturity	↑
European instead of American	↓
Counterparty Risk	↓
→ Cannot establish a bound	

Put-Call Parity (w/o Dividends)

$$C_t + PV(K) = S_t + P_t$$

- C_t = Current price of call
- $PV(K)$ = Present value of strike price (with risk-free rate)
- S_t = Current price of underlying
- P_t = Current price of put

Upper Bound (w/o Dividends)

• Payoff at maturity for put: $P_T = K - S_T$

→ Payoff maximized when $S_T = 0$: $P_T \leq K$

→ Put's highest value today is $PV(K)$: $P_t \leq PV(K)$

→ Using put's upper bound in PCP:

$$C_t = S_t - PV(K) + P_t \leq S_t$$

$$C_t \leq \$26.64$$

Upper bound for Citi trade = $\$26.64 \times 11.2\text{m} = \298m

Lower Bound (w/o Dividends)

- Put value cannot be negative: $P_t \geq 0$

- Using put's lower bound in PCP:

$$\begin{aligned}C_t &= S_t - PV(K) + P_t \\ &\geq S_t - PV(K)\end{aligned}$$

- Lower bound for one Citigroup call

$$\begin{aligned}C_t &\geq \$26.64 - \frac{\$26.64}{(1 + 2.28\%)^3} \\ &\geq \$1.74\end{aligned}$$

Lower bound for Citi trade = $\$1.74 \times 11.2\text{m} = \19.5m

Bounds: Summary

- Bounds cannot be derived from the listed options' prices
- Bounds can be established with PCP

Trade value bound	Lower	Upper
Not Accounting for Dividends	\$19.5m	\$298m
Price Paid		\$100m

- These (loose) bounds were not violated

Part 1: Conclusion

- The trades:
 - Actually simple: at-the-money 3-year calls
 - Risky:
 - + Leverage inherent to options
 - + Counterparty risk (for LIA only)
- Did Goldman “rip off” the LIA?
 - Beware hindsight bias: having lost money is no proof of a mistake
 - Cannot tell from listed options comparison and PCP
 - ➔ To assess the trade, an option pricing model is needed

Appendix B.1:

Simple PCP for Listed Options

Listed Options: PCP (w/o Dividends)

$$S_t + P_t - PV(K) = C_t$$

Type	Time to Mat.	Strike (\$)	Bid-Ask Avg. (\$)
Call	1	25	4.70
Call	1	30	2.61
Put	1	25	3.43
Put	1	30	6.25
Call	2	25	6.25
Call	2	30	4.30
Put	2	25	4.83
Put	2	30	7.73

Stock (\$)	+	Put (\$)	-	PV(K) (\$)	=	Call (\$)
26.64	+	3.43	-	24.44	=	5.62
26.64	+	6.25	-	29.33	=	3.56

Deviation from listed
20%
37%

26.64	+	4.83	-	23.90	=	7.57
26.64	+	7.73	-	28.68	=	5.69

21%
32%

- Why the price deviations?
 - Listed options are American not European
 - Dividends reduce call prices and enhance put prices → Ignoring them makes calls look cheap relative to puts → Markets expect Citigroup to pay dividends

Appendix B.2: PCP with Dividends

Put-Call Parity (w/ Dividends)

$$C_t + PV(K) = S_t e^{-\delta \cdot T} + P_t$$

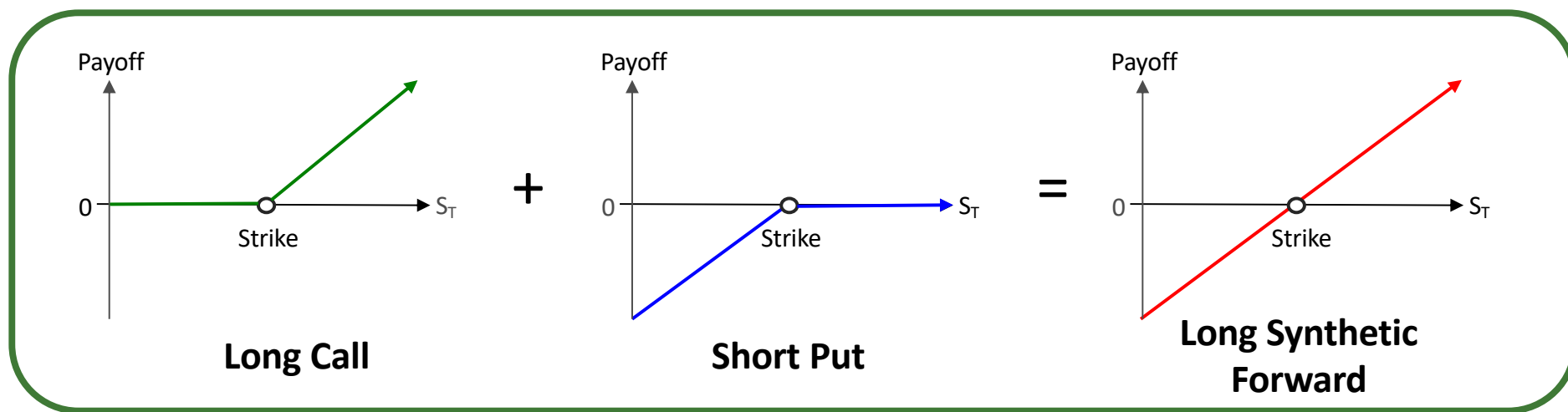
- C_t = Current price of call
- $PV(K)$ = Present value of strike price (with risk-free rate)
- S_t = Current price of underlying
- P_t = Current price of put
- δ = Expected dividend yield (annual dividend/stock price)
- T = Time to maturity (in years)

Citigroup's Historical Dividend Yield

- Citigroup's dividends = \$0.54/quarter over past 12 months
→ $\$0.54 \times 4 = \2.16 / year
- Current stock price = \$26.64
→ Annual dividend yield = $\$2.16 / \$26.64 = 8.1\%$

Ex-Dividend Date	Dividend per share
1-Nov-07	\$0.54
2-Aug-07	\$0.54
3-May-07	\$0.54
1-Feb-07	\$0.54
2-Nov-06	\$0.49
3-Aug-06	\$0.49

PCP with Dividends: “Proof”



- Long call and short put with strike equal to F = synthetic forward contract
- ➔ Must have a zero price
- ➔ Synthetic long forward with delivery price K has price of $PV(F-K)$:

$$C_t - P_t = PV(F - K)$$

- Forward price on dividend paying stock equals $FV(S_t \times e^{-\delta T})$ where FV denotes the forward value

$$\rightarrow C - P = PV(FV (S_t \times e^{-\delta T})) - PV(K) = S_t \times e^{-\delta T} - PV(K)$$

Listed Options: PCP (w/ Dividends)

Using $\delta = 8.1\%$ (historic yield)

$$S_t e^{-\delta T} + P_t - PV(K) = C_t$$

Type	Time to Mat.	Strike (\$)	Bid-Ask Avg. (\$)
Call	1	25	4.70
Call	1	30	2.61
Put	1	25	3.43
Put	1	30	6.25
Call	2	25	6.25
Call	2	30	4.30
Put	2	25	4.83
Put	2	30	7.73

Stoc k (\$)	+	Put (\$)	-	PV(K) (\$)	=	Call (\$)
24.56	+	3.43	-	24.44	=	3.55
24.56	+	6.25	-	29.33	=	1.48

Deviation from listed
25%
43%

22.65	+	4.83	-	23.90	=	3.58
22.65	+	7.73	-	28.68	=	1.70

43%
60%

- Why the price deviations?
 - Listed options are American not European
 - Using too high a dividend yield makes calls look expensive relative to puts
 ➔ Markets expect Citigroup to pay a dividend yield well below 8.1%

Appendix B.3: PCP with Dividends for Listed (American) Options

PCP for American Options

- For American options, PCP only yields a range:

$$S_t e^{-\delta T} - K \leq C_t - P_t \leq S_t - PV(K)$$

- This relation holds for the listed options:

Time to Maturity	Strike (\$)	Call (Bid-Ask Avg.) (\$)	Put (Bid-Ask Avg.) (\$)	Lower Bound	C-P	Upper Bound
1	25	4.70	3.43	-4.11	1.27	2.20
1	30	2.61	6.25	-9.11	-3.64	-2.69
2	25	6.25	4.83	-4.11	1.42	2.74
2	30	4.30	7.73	-9.11	-3.43	-2.04

PCP-implied Dividend Yield Lower Bound

- PCP with dividends → Lower bound on expected dividend yield

$$C_t - P_t \geq S_t e^{-\delta T} - K \quad \Rightarrow \quad \delta \geq -\frac{1}{T} \ln\left(\frac{C+K-P}{S_t}\right)$$

Time to Maturity	Strike (\$)	Call Premium (Bid-Ask Avg.) (\$)	Put Premium (Bid-Ask Avg.) (\$)	Dividend Yield Lower Bound
1	25	4.70	3.43	1.38%
1	30	2.61	6.25	1.08%
2	25	6.25	4.83	0.41%
2	30	4.30	7.73	0.12%

Appendix B.4

Value Bounds with Dividends

PCP: Upper Bound with Dividends

- Same equation with an adjustment for expected dividends:

$$C_t = S_t e^{-\delta T} + P_t - PV(K)$$

- Put's highest value today is $PV(K)$:

$$P_t \leq PV(K)$$

- Using put's upper bound in PCP:

$$C_t = S_t e^{-\delta T} - PV(K) + P_t \leq S_t e^{-\delta T}$$

$$C_t \leq 26.64 \times e^{-8.1\% \times 3} = \$20.89$$

Upper bound for Citi trade = $\$20.89 \times 11.2\text{m} = \234m

PCP: Lower Bound (w/ Dividends)

- Again, a put's price cannot be negative so PCP implies:

$$\begin{aligned}C_t &= S_t e^{-\delta T} - PV(K) + P_t \\ &\geq S_t e^{-\delta T} - PV(K)\end{aligned}$$

- Lower Bound for one Citigroup call:

$$C_t \geq S_t e^{-\delta T} - PV_t(K) = \$20.89 - \frac{\$26.64}{(1 + 2.28\%)^3} = -\$4.01$$

- In fact the lower bound is \$0

Lower bound for Citi trade = \$0 x 11.2m = \$0m

Bounds: Summary

- Bounds cannot be derived from the listed options' prices
- Bounds can be established with PCP

Trade value bound	Lower	Upper
Not Accounting for Dividends	\$19.5m	\$298m
Accounting for Dividends (8.1% yield)	\$0	\$234m
Price Paid	\$100m	

- These (loose) bounds were not violated

Appendix C

The EdF Trades

Payoff Diagram: The EdF Trades

